PROJECT DESIGN PHASE-II SPECIFICATION

StaffSync HRMS — Technology Stack & Architecture

3-tier enterprise cloud architecture specifications, dependencies, and operational characteristics.

Project Name	**StaffSync — Enterprise Human Resource Management System**
Architecture Pattern	Decoupled 3-Tier Enterprise Cloud (MVC + SPA)
Version & Author	v1.0.0 (Production Verified) | Parikshit Sharma

1. Technical Architecture Overview

The StaffSync platform operates as a modern decoupled 3-tier cloud enterprise architecture ensuring maximum isolation, developer velocity, and horizontal scalability:

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[ Client Presentation Tier ] ── HTTPS / Bearer JWT ──> [ Application Logic Tier ] ── Mongoose / TLS ──> [ Data Persistence Tier ]
• React 19 + TypeScript                    • Node.js 20+ & Express 4                    • MongoDB Atlas Cloud
• Vite 8 + Tailwind CSS v4                 • JWT & Bcrypt Middleware                    • Collections: users, employees,
• Vercel Global Edge CDN                   • Render Web Service                         leaves, attendance, training
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2. Table-1: Components & Technologies

S.No	Component	Description	Technology & Version
1	**User Interface**	Web-based responsive portal for HR, Managers, and Employees.	React 19, TypeScript, Vite 8, Tailwind CSS v4, Three.js
2	**Application Logic — Core**	Authentication, employee directory, leave management, and attendance.	Node.js 20+, Express 4, bcryptjs 2.4, jsonwebtoken 9.0
3	**Application Logic — Advanced**	Training curricula curation, priority bulletins, and executive analytics.	Express REST Router, cookie-parser, dynamic CORS
4	**Data Storage Layer**	Cloud document database storing credentials, dossiers, leaves, logs.	MongoDB Atlas Cloud Cluster, Mongoose ODM 8.24
5	**Cloud Hosting & CDN**	Production hosting with edge caching, automated SSL, health checks.	Vercel (Client) + Render (REST API)
6	**CI/CD Automation**	Automated validation pipeline running linting, syntax, and builds.	GitHub Actions (.github/workflows/ci-cd.yml)

3. Table-2: Application Characteristics

S.No	Characteristic	Description	Implementation Technology
1	**Modern Open-Source**	Unified JavaScript/TypeScript stack enabling high developer velocity.	React 19, Node.js, Express, Tailwind v4
2	**Scalable 3-Tier MVC**	Decoupled client-server architecture with RESTful contracts.	MVC Pattern, Express Controllers
3	**Strict Anti-Mock Security**	100% dynamic cloud persistence with zero static fallback stubs.	MongoDB Atlas cloud collections
4	**Cloud Cold-Start Resilience**	Calibrated 60s request timeout accommodating Render wakeups.	Client-side AbortController in apiClient.js
5	**Editorial Aesthetics**	Warm Ivory (#EAE9DF) and Evergreen (#173D32) design system.	Semantic Tailwind tokens, DM Serif Display
6	**Role Parity Empowerment**	HR and Managers both empowered to curate training programs.	authorizeRoles('hr', 'manager')